EVENTUAL PERIODICITY OF LINEAR GENERALIZED STONE–WALES PATHS IN AFFINE FULLERENE FAMILIES WITH CONVEX LATTICE-FACE DECOMPOSITIONS

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Abstract. We study literal linear generalized Stone–Wales paths in fullerene duals. Our main result concerns an integer affine family admitting a fixed decomposition into embedded convex lattice polygons with sides in the six triangular-lattice directions. With fixed gluing and affine corner coordinates, we give a uniform reduction of literal-path existence to a fixed monadic second-order sentence over unary finite words, using maximal face passages, their two-row outputs and all vertex collisions. Eventual periodicity on every eventually valid integer affine ray then follows from the known equivalence between MSO and regular languages. This is a geometric definability reduction, not a new general periodicity principle. We record the resulting ray statements for eleven specified catalogue families, alongside closed conditions on their stated domains. Existing Lean results establish arithmetic periodicity of several family predicates and determine two example rays for every ray parameter. For K3, C1 and O2, stated-domain graph classifications and actual-path periodicity on positive integer affine rays are Lean-proved on explicit quotient specifications, without the general prose bridge. Parameter-one rays are included; the O2 absence domain is $t, d \geq 2$. The separate C1 and O2 IPR conditions are outside Lean. The general eventual-periodicity statement is now proved in Lean against a specification written by the coordinator before the proof. Its correspondence to the geometric hypotheses is an explicit prose lemma, not a Lean theorem. The proof was supplied by Claude workers; no independent Lean proof by the research team is claimed. The separate R004 prose derivation is retained. Prior owner confirmation of the general specification is retained as a historical record, not a new verification procedure. Human proof peer review and novelty remain open; no executable geometric-to-MSO compiler or period algorithm is supplied.

Editorial and evidence status. This English edition continues Japanese v3, with convex-face Theorem 1a and catalogue Corollary 1b as its main results. The fixed-oblique-direction extension remains a proof sketch outside the abstract's claims. Section 9.3 separates the geometric reduction from known MSO, parametric Presburger and fixed-boundary periodicity results. R004 derived the convex-face proof without reading the research team's proof, construction or code; the statement was known. The O2 mapping metadata and D12 single-point disclosures remain: strict blindness is not claimed. One AI coordinator reviewed the route. Translation and packaging supply neither a third derivation nor human peer review. The coordinator-written Theorem 1a specification has a Lean proof and saved six-kernel acceptance. On 20 October 2026, the owner read version 2 with the Japanese guide and addendum and confirmed its statement meaning, not its proof or prose correspondence. Earlier authorship, affiliation, licensing and statement records remain factual; owner statement-meaning checks ended as a procedure that day. K3, C1 and O2 correspondence instead uses prior specifications, independent AI statement readers and finite comparisons. No new owner-confirmation procedure is added. A3's classification and 816-class census belong to the separate first paper. A new D12 classification is not incorporated.

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1. Definitions, signatures and valid rays

1.1. The literal path.

Definition 1. In the dual G of a fullerene, a literal linear gSW path is a sequence $p_0, \dots, p_e$ with odd $e \geq 3$, all vertices distinct, consecutive and distance-two vertices adjacent, $\deg p_0 = \deg p_e = 5$, and $\deg p_1 = \deg p_{e-1} = 6$.

Machine-readable claim: `def:literal`.

This is Definition 1(a) of [2]. Internal degree-five vertices are allowed. Even vertex count is part of the definition, not a new obstruction. Positive means existence of such a path; no chemical barrier, radial operation, or operation permitting heptagons is inferred. A fullerene is a simple three-connected cubic plane graph with only pentagonal and hexagonal faces. Its dual is a spherical triangulation with twelve degree-five vertices and all other degrees six.

When every three-cycle is facial and an edge has just the two common neighbours supplied by its incident triangles, an oriented triangle (a, b, c) has a unique next vertex: the common neighbour of b, c different from a. A simple path is a prefix of this forced sequence. If ρ is its first repeated-vertex index, a legal endpoint must occur strictly before ρ, not at ρ.
Machine-readable claim: `prop:forced`.

The conditional finite-graph statement is formalized as `R002.finite_literal_path_decision`. Its hypotheses and fixed start must be supplied by geometry. The noncomputable construction is not a certified executable Python or C classifier. For fullerene duals, the nonfacial-three-cycle exclusion in the received prose route uses the established cyclic edge-connectivity result cited in [4]; the original proof of that external result was not rechecked by the editor.

1.2. Signatures and eventual validity.

Machine-readable claim: `def:signature`.

Work in the triangular lattice $\Lambda = \mathbb{Z}^2$, with $R(x, y) = (-y, x + y)$. A signature specifies a fixed finite set of polygon charts, paired boundary segments, corner equivalence classes, and transition maps $z \mapsto R^j z + d$. Corner coordinates and translations are integer affine functions of the parameters; rotations and gluing combinatorics are fixed. Along a ray $p(t) = at + b$, $t \in \mathbb{N}$, we require eventually positive edge lengths, embedded chart interiors, only the specified seam identifications, noncrossing macro edges, and a valid simple fullerene dual. Every cone has a marked corner label. Seams need an ownership convention, and full corner equivalence classes are used. Positive parameters and positive area alone do not prove validity.

A component $a_i t + b_i$ eventually lies in the positive integers exactly when $a_i > 0$ or $a_i = 0, b_i \geq 1$. In the first case an entry bound is $\max(0, \lceil (1 - b_i)/a_i \rceil)$. Replacing 1 by 2 gives the analogous bound for a lower limit two. A negative slope has no infinite positive tail. These arithmetic entry bounds do not certify the quotient.

2. The main result: convex lattice-face decompositions

Theorem 1a (Convex lattice-face rays)**.** *For all sufficiently large integers t, suppose a valid simple fullerene dual has a decomposition into a fixed finite number of embedded closed convex lattice-polygon disks. Every side has positive length and one of the six lattice directions. Straight angles and auxiliary degree-six corners are permitted. No distinct boundary points of one disk are identified. Corner coordinates and transition translations are integer affine in t, while rotations, side pairings and corner identifications are fixed. All degree-five vertices occur among the marked corners, with a fixed label set. Then the set of t for which a literal path exists is eventually periodic. The conclusion also holds when these hypotheses hold separately on finitely many residue classes modulo a fixed integer.*

Machine-readable claim: `thm:aligned`.

Thus some $t_0 \geq 0, P \geq 1$ satisfy the same decision at t and $t+P$ for all $t \geq t_0$. The received theorem proves existence. No practical threshold, least period or implemented geometric-to-MSO/DFA compiler is supplied. The specification-level existence theorem is now Lean-proved; the prose correspondence lemma below transfers it to these geometric hypotheses. Section 3 retains the independent prose route; its original proof, corrections and coordinator review are fixed in the packet. The hypotheses describe the objects to which it applies, not a generic validity-testing algorithm.
Prose correspondence lemma (not formalized in Lean). The geometric hypotheses of Theorem 1a determine a fixed `ClaudeChallenge2.Sig` S and a tail on which `Valid` $S\,t$ and `TrianglesInCharts` $S\,t$ hold; the quotient `graph` $S\,t$ is the stated fullerene dual and `LiteralPath` is Definition 1.

Prose justification. For a convex lattice polygon with the six permitted side directions, its supporting lines give exactly three closed strips $L_x \leq x \leq U_x$, $L_y \leq y \leq U_y$, $L_s \leq x+y \leq U_s$. Their extrema are selected from finitely many affine corner coordinates; after their orders stabilize, all six bounds are integer affine. Retain the fixed charts, rotation exponents, affine translations and marked corners as the fields of S. For each paired side, use its prescribed transition as a gluing record.

No extra identifications from unrestricted records. Let g unfold a paired segment $A \subset K$ onto $g(A) \subset K'$, and let ℓ be its supporting line. The consistently oriented disks $g(K)$ and K' occupy opposite closed supporting halfplanes, so $g(K) \cap K' \subset \ell$. Their overlap there is an interval I containing $g(A)$. If I continues beyond an endpoint of $g(A)$, both supporting edges continue straight at that point. The two disks each contribute angle π. A fullerene dual has total cone angle $5\pi/3$ or 2π, so the point is flat, the two disks exhaust its neighbourhood, and their actual seam must continue across it. The orientation-preserving lattice isometry along the continued seam is the same g: it agrees on a nontrivial line segment. Repeat across the finitely many side subdivisions until reaching the endpoint of I. Thus every pair identified by this unrestricted record is truly the same surface point, even if it lies beyond the named subdivision. Conversely, the prescribed paired sides are included in the records. A self-pairing of distinct points is excluded by the embedded-closed-disk hypothesis; an identity record is harmless. Transitive closure therefore gives exactly the prescribed seam and full corner equivalences, without extra identifications. This supporting-line and cone-angle argument is prose, not a theorem certified by the Lean kernel. Thus sites modulo these records are precisely the surface vertices, and common-chart unit steps are precisely the graph edges. Closed-disk embeddedness supplies the injectivity clause of `Valid`; the fullerene degrees and marked-corner hypothesis supply its degree clauses.

Use the same external fullerene fact as in Section 1: all dual three-cycles are facial, by cyclic edge-connectivity [4]. Each edge therefore has exactly the two distinct opposite vertices of its incident triangular faces as common neighbours. This gives `TwoCommonNeighbors`. Since chart boundaries follow lattice edges, no open unit triangle is cut by a boundary; the covering decomposition puts every triangular face inside a closed chart. Every graph triangle is facial, so its three vertices lift there with all three differences unit vectors. This is exactly `TrianglesInCharts`. These geometric identifications and the external fullerene fact are part of this prose bridge, not additional Lean-proved declarations or a fresh audit of the cited proof.

Apply `ClaudeChallenge2.main` on the resulting tail. For fixed residue branches write $t = r + qn$ and apply the construction to each branch. The received Lean declaration `ClaudeChallenge2.main_residue` then combines the branches in the original parameter, assuming eventual agreement of the decision with each branch's `PathExists`. That agreement is supplied by this prose representation lemma; the combination of eventual periods is Lean-proved. The owner confirmed the version-2 specification's meaning on 20 October 2026; the check is recorded in `OWNER-STATEMENT-CHECK.md`. It is not proof peer review or review of this prose correspondence lemma.

Corollary 1b (Specified catalogue families)**.** *On every integer affine ray eventually remaining in the valid positive integer domain of each of A1, A2, B2, C1, K3, E10, D12, D14, D1 and O2,*

literal-path existence is eventually periodic. The same conclusion holds for A3 on rays eventually satisfying $k > l > 0$, by its separate classification and arithmetic argument [1].

Machine-readable claim: `cor:catalog`.

There are eleven families in this statement. Positive boundaries with a parameter fixed at 1 are included, notably K3 and O2 and E10 with $r = 1$. Collapsed zero boundaries are not. The corollary gives no two-variable closed classification merely from ray periodicity. The K3, C1 and O2 parts now also have direct Lean ray proofs on their quotient graphs, including positive boundary parameters, without invoking the general prose correspondence lemma. Their ray declarations start with positive natural coordinates and nonnegative slopes; an eventually positive integer ray is covered by reindexing its positive tail. The other family routes retain their stated scope. A3 generally has changing parent-edge directions and is not silently placed under Theorem 1a. The catalogue terminology and corrections are Graver's [3]; the constructions are not claimed as new. Additional leapfrog families are discussed only as prospects in Section 8.

3. Proof of the convex-face theorem

This section presents R004's second derivation on the common six-direction domain. It does not use the uncompleted oblique-direction transfer conditions. Its arithmetic-to-word reduction is specific; arbitrary Presburger relations are not claimed to be finite-word MSO.

3.1. **Maximal face passages and their outputs.** *Machine-readable claim:* `lem:rows`.

Rotate a passage direction to $(1, 0)$. A suitable phase of the forced vertex sequence is

$$q_{2j} = (j, h), \qquad q_{2j+1} = (j, h + 1). \tag{1}$$

A convex six-direction polygon is specified by

$$L_x \le x \le U_x, \qquad L_y \le y \le U_y, \qquad L_s \le x + y \le U_s.$$

Its bounds are affine in t after the finitely many max/min orders stabilize. Redundant collinear sides do not alter this description. With $L_y \le h \le U_y - 1$, the triangle $T_i = \{q_i, q_{i+1}, q_{i+2}\}$ lies in the polygon precisely in the ranges

$$\begin{aligned} A_e &= \max(L_x, L_s - h), & B_e &= \min(U_x - 1, U_s - h - 1), \\ A_o &= \max(L_x, L_s - h - 1), & B_o &= \min(U_x - 1, U_s - h - 2). \end{aligned} \tag{2}$$

Even indices are $2j$ for $A_e \le j \le B_e$, odd indices are $2j + 1$ for $A_o \le j \le B_o$. Put $\delta = A_e - A_o$ and $\epsilon = B_o - B_e + 1$, both in $\{0, 1\}$. The full triangle-index set is empty or the interval

$$\ell = 2A_e - \delta \le i \le u = 2B_e + \epsilon. \tag{3}$$

This includes the one-odd-triangle case $A_e = B_e + 1$, $\delta = \epsilon = 1$. It must not be discarded as an empty even range.

The first two vertices of an entered passage already belong to the previous passage. Its new output is only $q_{\ell+2}, \dots, q_{u+2}$, comprising

$$\begin{aligned} y = h &: \quad A_e + 1 \le x \le B_e + 1, \\ y = h + 1 &: \quad A_e + 1 - \delta \le x \le B_e + \epsilon. \end{aligned} \tag{4}$$

For example the upper row has $x \ge L_x + 1 - \delta$, $x + y \ge L_s + 2 - \delta$, $x \le U_x - 1 + \epsilon$, and $x + y \le U_s + \epsilon$. The last constant was corrected by hand in the received proof, followed by a new finite check; the check did not originally discover that error. Empty rows are allowed, but a nonempty passage has nonempty output.

Root and terminal passages can be truncated. There are finitely many marked cone corners and at most ten oriented initial triangles at each. The initial triangle fixes h and an index of the form twice an affine expression plus fixed parity. Store its first two vertices separately. An endpoint corner and choice of output row similarly determine the last index. Truncation adds affine bounds

on x. Both cuts are made when the whole path lies in one passage. Internal degree-five corners are not compulsory stopping points. The two endpoint neighbours must be distinct from all degree-five labels.

3.2. **States, transitions and all collisions.** *Machine-readable claim:* `lem:restricted`.

A state is (τ, h), where the finite type τ records face, direction, max/min branches, parity, subdivided entry/exit side, and root/terminal information. A constant C bounds $|h| \le Ct + C$. The number of states or face passages need not be bounded independently of t. The relevant relations are Boolean combinations of

$$\pm h \le at + b, \qquad \pm h \pm h' \le at + b, \qquad at + b \ge 0, \tag{5}$$

and fixed-modulus congruences. A relation such as $2h + h' \le at + b$ with distinct free variables is not part of this language. Substituting one variable for another can be reduced by fixed parity and rounding.

Within a fixed type, each endpoint's three normal coordinates has the form $c(t)$, $h + c(t)$ or $-h + c(t)$. When the last ordered pair of a passage is glued to the next one's first ordered pair, the directions are rotated together. In direction-aligned coordinates the linear part of this transition is the identity, so $h' = h + c(t)$, not a doubling map. The paired unit edge determines the opposite face; its interior contains no macro corner. The conditions therefore have form (5).

Each output row fixes one of $x, y, x + y$ at $\pm h + c(t)$ and imposes affine bounds in all three normals. For distinct row normals the determinant is ± 1: their intersection is integral, and the third normal is their sum or difference. Substitution into both rows gives (5). Equal normals first require equal values; the remaining integer interval is nonempty exactly when every lower bound is at most every upper bound. The same restricted form results. Between different faces, compare through every shared side and every common marked corner label. Rotations permute the three normals with signs. Embeddedness and fixed gluing ensure these exhaust quotient identifications, even when faces share several sides. Auxiliary flat corners are included. Comparison with the root's two saved vertices uses the same formulas. Since an embedded face row is injective, pairwise disjoint outputs and their disjointness from the root are exactly the full vertex-distinctness condition.

3.3. **Folding the restricted arithmetic into a finite word.** *Machine-readable claim:* `lem:folding`.

For $t \ge 2$, write a height variable uniquely as $h = jt + y$, $0 \le y < t$. The quotient j ranges over a fixed finite set. Set $t = 2n + \eta$, $\eta \in \{0, 1\}$, and split y into

$$L : y = v, \qquad R : y = t - 1 - v \quad (0 \le v < n), \qquad M : y = n \quad (\eta = 1). \tag{6}$$

Original type, j and $L/R/M$ are finitely many copies of the positions of a unary word of length n. A middle copy is placed at the first position and interpreted as coordinate n, independent of that position.

For two ordinary positions the inequalities become

$$\alpha v + \beta w \le 2cn + d, \qquad \alpha, \beta \in \{-1, 1\}.$$

The coefficient of n is even. If $\alpha = -\beta$, then for $c \ne 0$ truth stabilizes, while $c = 0$ is order with a fixed successor offset. If $\alpha = \beta = 1$, the sum lies between 0 and $2n - 2$. The only nontrivial cases are $c = 0$, a finite list near the left end, and $c = 1$. For $c = 1$, putting $a = n - 1 - v, b = n - 1 - w$ gives $a + b \ge -d - 2$, either automatic or the exclusion of finitely many near-end pairs. The case $\alpha = \beta = -1$ is its integer complement. If one position is a middle copy, only one variable remains and the nontrivial inequalities are fixed distances from an end. Two middle copies give an affine truth value that stabilizes.

Fixed-modulus congruences are represented by cyclic MSO colours on the word: the first position has colour zero and each successor increments the colour. The final colour determines n modulo the fixed modulus. Finite copies of element or set quantifiers are expanded copy by copy. Thus

every fixed Boolean combination used above is MSO-definable. This argument does not permit an arbitrary Presburger relation such as $v + w \le n$; it uses the even coefficient obtained by the folding step.

3.4. **The exact path sentence and unary regularity.** *Machine-readable claim:* `lem:mso-path`.

Choose a nonempty state set S, root a and terminal b. Terminal states have no outgoing transition and root states no incoming one. Record the parity of the number of vertices already output as a finite type. Initially it is even, since the root has two saved vertices; each transition adds the parity of $u - \ell + 1$. Impose the following MSO conditions:

(1) If $a = b$, $S = \{a\}$. Otherwise a has in/out degrees $(0, 1)$, b has $(1, 0)$, and every other state has $(1, 1)$ within S.
(2) Every nonempty proper subset of S has a transition, in either direction, across its boundary. This is unary-set connectivity.
(3) Outputs of distinct states are disjoint, and all outputs are disjoint from the two saved root vertices.
(4) Endpoint degrees are $5, 6, 6, 5$ as in Definition 1. Total vertex count is even and at least four.

For one state, the last size condition is $u - \ell + 1 \ge 2$; with two or more nonempty-output states it is automatic. Degree conditions are membership or nonmembership in the finite marked-cone set. No binary relation quantifier is used.

On a finite set, the first two conditions give one directed path through all of S: a reached cycle would create an extra incoming edge, and a separate cycle would contradict connectivity. Join the passages in this order. Ordered seam pairs give consecutive and skip adjacency; the third condition gives distinct vertices. Size, parity and degree conditions give exactly the literal path.

Conversely, split any literal path into maximal consecutive passages in its convex faces, truncating only at its ends. Equations (2)–(4) recover their states. Nonempty output and path simplicity prevent a repeated state. Any extra eligible successor within S would share the next output vertex with the actual successor, contradicting disjointness. Hence the selected states satisfy all four conditions. One-passage paths and paths with internal cones are included in both directions.

Finite-word MSO defines exactly the regular languages [5, Section 3.1, Theorem 3.2]. Apply this established result only at this last logical step. For each η, a deterministic automaton over one letter eventually repeats a state, so its accepted lengths n are eventually periodic. Combine $t = 2n + \eta$ and, if necessary, the initially fixed residue branches. This proves Theorem 1a. The geometric reduction belongs to the received R004 proof, not to the cited automata theorem. Neither the full proof of that external theorem nor an MSO compiler was newly verified by this manuscript's editor.

4. Applying the theorem to the specified families

Machine-readable claim: `prop:family-decompositions`.

The applicability evidence is R004's fixed family-decomposition report and its coordinator review. D12 uses two subdivided hexagons and D14 two convex hexagons whose long sides are subdivided with lengths r, s, r. O2 uses two hexagons and six triangles. K3 uses two caps, each divided into four triangles, and three hexagons. These decompositions retain the positive parameter-1 boundaries.

D1's original thirty-degree sides do not directly satisfy the six direction hypothesis. Cut a triangle $A_iB_iW_i$ from each oblique side and move it to the back by the same lattice isometry. For example,

$$A_0 = (r, 0), \quad B_0 = (r + p, p), \quad W_0 = (r, p), \quad C = (-p, r + 2p),$$

and $W_0 - C = \alpha(A_0 - C) + \alpha(B_0 - C)$ with $\alpha = (r + p)/(2r + 3p)$ and $0 < 2\alpha < 1$. The pieces are contained and disjoint. The result is eight convex pieces: one regular hexagon, six quadrilaterals, and the back hexagon. New corners are flat degree six. Area equality alone is not used to infer the same quotient.

For E10 put $m = 2r + s$, tube circumference $C = 3m$. Move the longitudinal cut from (p, p) to $(0, 2p)$, obtaining three lattice parallelograms of width $m < C$ and two caps: five disks, also when $r = 1$. Cut-corner positions involve $p \bmod C$. If C grows, $\lfloor p/C \rfloor$ eventually stabilizes; if C is constant, use fixed residue classes. Delete duplicate auxiliary corners and fix their order to obtain affine coordinates and fixed gluing. This does not classify E10 with $r = 1$.

Family and domain	Received reason for all valid affine rays
A1, $r \geq 1$	Integer GC absence, including the separate $r = 1$ case; eventual period one.
A2, $p \geq 1$	Nonconstant rays eventually give nonprimitive GC images; constant rays are constant graphs.
B2, $p, r \geq 1$	Its closed condition is a finite disjunction of affine equalities.
C1, $a, b \geq 1$	Lean graph iff $b = 1$ and actual-path ray periodicity; no general prose bridge.
A3, $k > l > 0$	Separate complete classification and arithmetic; Section 6.
K3, $r, s \geq 1$	Lean literal-path ray periodicity, including parameter 1; no general prose bridge is used.
E10, $p, r, s \geq 1$	Arithmetic for $r \geq 2$; the five-piece decomposition includes $r = 1$.
D12, D14, D1	The convex decompositions on valid positive integer parameters.
O2, $t, d \geq 1$	Lean actual-path ray periodicity, including parameter 1; Lean absence for $t, d \geq 2$. No general prose bridge.

This proves Corollary 1b using the stated geometric receipts and the separate arithmetic results. It does not cover all 112 catalogue entries or every zero-boundary degeneration.

5. Closed conditions and the scope of partial results

The following are received classifications of valid catalogue graphs. The R004 prose arguments are retained; K3, C1 and O2 now additionally have Lean graph classifications on their respective stated domains. Independent derivation of the same conclusion is distinct from checking every coordinate convention in the research team's tables. Lean arithmetic for a predicate does not by itself prove its equivalence to a graph property.

5.1. **The specified closed domains.** *Machine-readable claim:* `prop:closed`.

Family/domain	Literal path condition	Received scope
A1, $r \geq 1$	Never positive	Integer GC for $r \geq 2$; at 1 no degree-six vertex. R004 report 4 and GC proof.
A2, $p \geq 1$	$p = 1$	Existing C_{60} witness and integer GC. R004 report 14 does not supply a new $p = 1$ witness.
C1, $a, b \geq 1$	$b = 1$; IPR iff $a, b \geq 2$	R004 prose proof retained; Lean graph iff received for paths only. The IPR condition is outside Lean.
B2, $p, r \geq 1$	$p = 1$ or $\lvert p - r \rvert = 1$	R004 report 4 and B2 proof, whole positive domain.
K3, $r, s \geq 2$	$s \equiv \pm 1 \pmod{r}$	R004 positive and necessary prose proofs retained; full Lean graph iff now received, without a prose bridge.
O2, $t, d \geq 2$	Never positive	R004 prose absence retained; Lean graph absence received, without a prose bridge. IPR remains outside Lean.
E10, $p, s \geq 1, r \geq 2$	$m = 2r + s$ and $p \equiv d \pm 1 \pmod{m}$ for some $d \in \{0, r, r + s\}$	R004 reports 5–6 and both prose directions. Predicate-ray arithmetic formalized; full graph iff not formalized.

K3 writes $s = qr + u$: $u = 0$ is excluded by integer GC, $2 \leq u \leq r - 2$ by belts of widths $r, u, r - u$, and $u = 1, r - 1$ by explicit simple paths. The overlapping small cases $r = 2, 3$ are separate. The iff is not extended to $r = 1$ or $s = 1$. E10's candidate congruences come from the positions of corners around the tube; reaching a cone does not alone establish simplicity. The research team's full table of eighteen E10 regions, walls and witnesses is not all independently reviewed.

5.2. **Partial branches and boundary examples.** *Machine-readable claim:* `prop:partial`.

For O2, absence on $t, d \geq 2$ now also has a Lean graph proof, while IPR on that domain remains a received prose result. The mapping-metadata disclosure is retained. Boundary paths for $O2(1, d)$ with $d \geq 3$ and $O2(t, 1)$ with $t \geq 3$ are Lean-proved; existence at $O2(1, 1)$, $O2(1, 2)$ and $O2(2, 1)$ is outside the received Lean results. The ray theorem nevertheless covers these constant rays without deciding their path existence. For D14 and D1 with both parameters at least two, the research team has belt-based absence drafts; R004's all-ray periodicity receipt is not a new independent proof of that closed absence claim. Common integer scaling by at least two gives absence in the specified leapfrog domains, but does not classify their primitive points.

For $E10(p, 1, s)$, $m = s + 2$, the research team's witnesses give sufficiency when $p < m$ or $p \bmod m \in \{0, 1, 2, m - 2, m - 1\}$. General necessity is not proved. Among the 84 positive (p, s) with $N = 12 + 2s^2 + 12s + 12p(2 + s) \leq 740$, exactly $E10(9, 1, 4) = C_{740}$ is negative; all 83 smaller examples are positive. $E10(17, 1, 11)$ and $(31, 1, 7)$ give further negatives at C_{3038}, C_{3542}. These three examples are non-IPR. R004's report 4 independently constructed and checked them. This is minimality only inside the $r = 1$ E10 family, not a general iff.

The pre-existing D12 branches are retained as historical scope: $\gcd(r, s) \geq 2$ gives absence; $r \geq 3$, $s = qr \pm 1$, $q \geq 1$ gives existence by the separately derived simple witnesses. The research team's $\lvert r - 3s \rvert = 1$ and $r \geq 4s$ branches have their earlier review limitations. A full D12 classification, including newer work, is outside this paper and this packet.

6. Arithmetic on affine rays and exact examples

6.1. **The determinant reduction.** *Machine-readable claim:* `lem:divisibility`.

Let $A(t) = \alpha t + \beta$ be eventually positive and $B(t) = \gamma t + \delta$. A constant A gives a fixed-modulus congruence. If $\alpha > 0$, then

$$A \mid B \quad \Longrightarrow \quad A \mid \alpha\delta - \gamma\beta. \tag{7}$$

For nonzero right side, divisibility is impossible once $A > |\alpha\delta - \gamma\beta|$. For zero right side, the ratio $B/A = \gamma/\alpha$ is constant and is either always integral or never integral. Finite Boolean combinations of affine divisibilities are eventually periodic; a growing modulus gives eventual constancy. Take a common multiple of fixed moduli and a maximum stabilization threshold. This elementary arithmetic is not proposed as new number theory. The received Lean files establish the exact predicates below.

For all integer ray coefficients, define parameters by $(at + b).\,\mathrm{toNat}$. The formal statements for all rays use *domain and positive predicate*; they do not call out-of-domain graphs positive. On an eventually positive domain the conversion agrees with the integer affine parameter.

Machine-readable claim: `thm:k3-arithmetic`.

`Claude2.k3_ray_all` proves eventual periodicity of K3Dom $\wedge$ K3Pos on all integer affine rays, where the domain is $r, s \geq 2$ and the predicate is $r \mid (s-1)$ or $r \mid (s+1)$. If r is nonconstant on a valid tail, `k3_ray_const` gives eventual constancy. If $r = b \geq 2$ is constant, `k3_ray_period` gives period b from any specified domain-entry threshold. These need not be least periods.

Machine-readable claim: `thm:e10-arithmetic`.

`Claude2.e10_ray_all` proves the corresponding all-ray statement for domain $p, s \geq 1, r \geq 2$ and the six divisibilities in the E10 condition. If $2r + s$ is nonconstant, the predicate is eventually constant; when it is constant, `e10_ray_period` gives that positive modulus as a period on the given domain tail.

Machine-readable claim: `thm:other-arithmetic`.

`b2_ray_all`, `c1_ray_all`, `a1_ray_all` and `a2_ray_all` give eventual constancy of their domain-qualified predicates. The conjunction/disjunction/negation closure and domain qualification are proved in the same arithmetic layer.

6.2. **A3 is a separate variable-direction case.** *Machine-readable claim:* `thm:a3-arithmetic`.

For $k(t) = at + b, l(t) = ct + d$ set $\Delta = ad - bc$. If $\Delta \neq 0$, $\gcd(k, l) = \gcd(k, l, \Delta)$, so primitivity is periodic modulo $|\Delta|$. When $\Delta = 0$ and the ray is nonconstant, it is eventually an increasing integer multiple of a fixed primitive direction and hence negative by integer GC. Constant rays are constant. In the primitive case put $m = \min(l, k - l)$. The A3 criterion is $m = 1$, or quotient $q = \lfloor k/m \rfloor \geq 3$ with $m \mid (k \pm 1)$, or $q = 2$ with $r = k - 2m = 1$ or $r \mid (m \pm 1)$. Branches and quotient stabilize, including an integer limiting ratio by checking the sign of the constant remainder. The persistent $q = 2, r = 0$ growing branch is not primitive. Apply (7) to the remaining tests.

The first paper proves the full graph classification. Its `ClaudeCor.criterion_ray`, restated as `Claude2.a3_ray`, formalizes the arithmetic existence of a threshold and positive period. A separate release bridge gives graph-path periodicity. Neither result certifies an executable threshold/period algorithm. R004's separate arithmetic argument permits period $|\Delta|$ for nonconstant $\Delta \neq 0$ rays and period 1 otherwise. This is not a least-period statement or formalization of the general geometric theorem. After classification, eventual periodicity is also a consequence of [6, Theorem 1.15] and is not a separate novelty claim.

6.3. **Four previously solved rays.** *Machine-readable claim:* `ex:exact-rays`.

Ray, $t \geq 0$	Positive values	Valid period and threshold
D12$(124+3t, 41+t)$	All t	$P = 1, t_0 = 0$
D12$(125+3t, 41+t)$	None	$P = 1, t_0 = 0$
K3$(43+2t, 170+5t)$	$40, 42$	$P = 1, t_0 = 43$
E10$(200+5t, 17+2t, 23+t)$	$3, 10, 17, 34$	$P = 1, t_0 = 35$

The original compressed-word calculations did not use the known family congruences. Later independent prose arguments recover all t from independently derived family conditions, or from explicit D12 blocks. The expected statements were known, not blind predictions.
Machine-readable claim: `ex:k3-formal`.

`Claude2.k3_example` now proves for every $t \in \mathbb{N}$ K3Pos$(43+2t, 170+5t) \leftrightarrow (t = 40 \vee t = 42)$.
Machine-readable claim: `ex:e10-formal`.

`Claude2.e10_example` proves the corresponding E10 predicate iff $t \in \{3, 10, 17, 34\}$. The determinant constants used are $123, 127$ for K3 and $142, 144, 739, 749, 681, 691$ for E10. Divisibility bounds t before the remaining cases are checked by kernel arithmetic. These are all-t arithmetic proofs, not graph iff proofs by themselves.

The earlier D12 two-ray receipt covers every $s \geq 41$: D12$(3s+2, s)$ is negative and D12$(3s+1, s)$ has a path with $56s + 12$ distinct vertices. Its reported computer-assisted proof checks 2,058 nonnegativity conditions and 756 no-integer-solution conditions, 2,814 in total. A separately coded reread by the same verifier is not a further reviewer. The coordinator's independent construction at $s = 58$ agrees, with a 3,260-vertex positive witness; it does not audit all 2,814 conditions. These records are neither Lean proofs nor a full D12 classification.

7. Received formalization and what it does not cover

7.1. **Theorem 1a on the coordinator's specification.** The current entry points are `ClaudeChallenge2.main` in `Formal/Claude3ComparatorSolution.lean` and `ClaudeChallenge2.State.main_statement` of type `ClaudeChallenge2.MainStatement` in `Formal/Claude3Thm1a.lean`. For every fixed signature S and t_1, they prove that if `Valid` $S\,t$ and `TrianglesInCharts` $S\,t$ hold for all $t \geq t_1$, then some t_0 and positive P satisfy `PathExists` $S\,t \leftrightarrow$ `PathExists` $S\,(t+P)$ for every $t \geq t_0$. The independent challenge was written by the coordinator before the proof and amended to version 2 to include `TrianglesInCharts`. The paper-to-specification passage is the prose lemma following Theorem 1a; it is not formalized. The Lean assumptions are weaker than the paper's geometric hypotheses: they do not require a sphere, exactly twelve cones, or nondegenerate polygon disks. With the same conclusion, the exact formal result is therefore a stronger sufficient theorem, not a literal restatement of the paper's geometric objects. `ClaudeChallenge2.main_residue` in `Formal/Claude3Residue.lean` proves the fixed-residue variant after the decision-to-branch agreements are supplied.

The proof was supplied by Claude workers, through finite-word MSO, height folding, chart geometry, quotient identifications, state reduction and sentence assembly. It is not an independent Lean proof by the research team. The research team's serial build and fresh axiom queries for all three declarations found only `propext`, `Classical.choice` and `Quot.sound`. The saved coordinator Comparator receipt reports six kernels accepting the proof of `ClaudeChallenge2.main` in paranoid mode; that receipt is not a separate six-kernel check of the residue declaration. This edit does not rerun external kernels or provide OS process isolation. The owner's version-2 statement-meaning confirmation is recorded separately. Human proof peer review remains unrecorded. Neither this formalization nor R004's separate prose proof establishes novelty or an executable geometric compiler, practical threshold, or least period.

7.2. **K3 graph classification and ray periodicity.** *Machine-readable claim:* `thm:k3-sufficient`.

The new entry declarations are

```
Claude4K3.k3_statement : Claude2Aligned.K3Statement;
Claude4K3.noThree;
ClaudeChallengeK3.main.
```

For natural $r, s \geq 2$, the first proves the full iff between an actual literal path in `Claude2Aligned.K3Graph`(r, s) and $s \equiv \pm 1 \pmod{r}$. The second proves that no edge has three common neighbours, using a classification of graph triangles in the cap or tube. Together with the earlier necessity machinery, this supplies the missing necessity direction. R004's positive and necessary prose proofs remain separate evidence.

The third proves eventual periodicity of literal-path existence on every ray $(r, s) = (r_0 + at, s_0 + bt)$ with $r_0, s_0 \geq 1$ and $a, b \geq 0$. There are no `Valid` or `TrianglesInCharts` hypotheses. The K3 gap previously filled by the general prose correspondence lemma is therefore closed in Lean on this explicit quotient specification. The proof constructs the needed common-neighbour property directly and does not construct a signature isomorphism into Theorem 1a. Nonconstant parameter-one boundary rays use explicit eventual paths; constant rays are periodic because their decision never changes. The modular iff itself is still restricted to $r, s \geq 2$.

The specification uses eleven convex polygons, equivalence closure of explicit affine side gluings, and adjacency from unit differences inside a common chart. The literal path and actual neighbour-set degree definitions are those of the first paper's specification. The earlier sufficient declaration `Claude2K3.k3_sufficient` of type `Claude2Aligned.K3Sufficient` is retained.

The proof follows R004's tube construction. A glued-site invariant is $(\mathrm{hgt}, \mathrm{ang} \bmod 6r)$. It is preserved by all 42 directed edge gluings. It is not globally injective: the two central cap triangles are not separated by it. On tube points with horizontal coordinate difference at most one, however, equality of quotient vertices forces equality of points. The explicit paths are

$$P_n^+ = (-(n \bmod 2), \lfloor (n+1)/2 \rfloor), \qquad P_n^- = (n \bmod 2, \lfloor n/2 \rfloor),$$

with $4s - 2$ vertices for $s = qr + 1$ and $4s + 4$ for $s = qr - 1$. Chart membership, consecutive and skip adjacency, distinctness, and the four endpoint degree requirements are proved on the quotient. The invariant is not used to claim a global vertex normal form.

Machine-readable claim: `lem:k3-corners`.

`Claude2K3.deg_corner` proves degree five for all twelve labelled corner representatives. It does not by itself prove they exhaust all degree-five vertices or all global vertex types.

Machine-readable claim: `lem:k3-hex-interior`.

`deg_hex_interior` proves degree six in hexagon interiors.

Machine-readable claim: `lem:k3-tri-interior`.

`deg_tri_interior` proves degree six in the six noncentral cap-triangle interiors. General seam interiors and the central cap triangles are not covered by those older partial degree lemmas alone. The new necessity proof uses the cap/tube triangle classification and common-neighbour bound; the K3 iff in Section 5 is now formalized. This does not relabel the partial degree lemmas as a global degree classification.

The older graph definitions were written by a Claude worker. Before the new proof, the coordinator copied them into a separate K3 challenge. Static comparison confirms the copied definitions and the version-2 challenge/solution statement agree. An independent AI reader of version 1 found the missing parameter-one boundaries; version 2 corrected them before the proof. The reader's finite checks and a separate Python mirror support the catalogue correspondence, without constituting a universal catalogue-isomorphism proof or human proof review. No owner-confirmation field or procedure is used for K3. Shared specification files contain unused D12 definitions, but no new D12 classification is a root or claimed result of this package.

7.3. **C1 graph classification and ray periodicity.** The two received entry declarations are

`ClaudeChallengeC1.classification`;
`ClaudeChallengeC1.main`.

On `Claude2Aligned.c1Graph`(r, s), natural $r, s \geq 1$ admit a literal path if and only if $s = 1$. Lean's r, s are the paper's a, b; this is the path condition $b = 1$ in Section 5, including $a = 1$. The separate IPR equivalence $a, b \geq 2$ is not stated or proved by these declarations. R004's C1 prose proof remains separate evidence. The second declaration proves actual-path eventual periodicity on every ray $(r_0 + ut, s_0 + vt)$ with $r_0, s_0 \geq 1$ and $u, v \geq 0$, including constant rays. Eventually positive integer rays are covered after shifting their starting index; this reindexing is explanatory arithmetic, not a separately received all-integer Lean declaration.

The eight-cell specification has four hexagons with alternating sides s, r, s, r, s, r and four triangles of side r. The vertex bijection `Claude4C1.vEquiv` and adjacency theorem `adj_iff` identify its quotient with the lattice surface in $\mathbb{Z}^4$ cut out by $0 \leq x_i \leq r + s$, $\sum_i x_i = 2r + s$, and some $x_i \in \{0, r + s\}$. The transfer `Claude4C1.c1_iff_g4` preserves literal-path existence. The proof establishes degrees and excludes three common neighbours of an edge directly, then uses forced strips for $s \geq 2$ and an explicit path for $s = 1$. It does not use the external cyclic edge-connectivity theorem of the R004 prose argument. C1 thus needs no general prose geometric-to-specification bridge for these path claims.

The coordinator fixed the specification and challenge before the proof. One independent AI statement reader returned "faithful with caveats", checking the parameter correspondence by the area formula $8r^2 + 16rs + 4s^2$ and identifying IPR as outside the challenge. The saved mirror agrees with the R004 construction on counts, degree sequence and the path decision for 36 pairs $1 \leq r, s \leq 6$; the reader's 25 finite pairs lie within that range and are not a second disjoint census. These finite checks are not a universal catalogue-isomorphism certificate or human proof review. No owner-confirmation field or procedure is added. The challenge has two intentional placeholders and is excluded from proof dependencies; the solution proves both statements. Exact sources and receipt hashes are supplied in the machine packet.

7.4. **O2 absence and ray periodicity.** The received declarations are

`ClaudeChallengeO2.classification`;
`ClaudeChallengeO2.main`.

The first proves that `Claude2Aligned.o2Graph`(t, d) has no literal path for natural $t, d \geq 2$. The second proves actual-path eventual periodicity on every ray $(t_0 + an, d_0 + bn)$ with $t_0, d_0 \geq 1$ and $a, b \geq 0$, including parameter-one and constant rays. The parameter names agree with the paper. Reindexing an eventually positive integer ray to its positive tail remains explanatory arithmetic. R004's prose absence proof is retained; these O2 path claims now need no general prose geometric-to-specification bridge. IPR is not formalized.

The eight-cell specification uses two hexagons with alternating sides $2t, d, 2t, d, 2t, d$, each long side split at its midpoint, and six triangles of side t. The equivalence `Claude5O2.o2_iff_gm` identifies it with two overlaid hexagons sewn along the three d-sides, with a rhombic pocket inserted into each $2t$-side slit. The proof directly rules out three common neighbours on an edge, locates degree-five vertices at the twelve corners, and covers the 120 starting triangles by ten closed strips. It does not use the external cyclic edge-connectivity theorem of the R004 prose argument.

Explicit boundary paths are proved for $(1, d)$ with $d \geq 3$ and $(t, 1)$ with $t \geq 3$. Path existence at $(1, 1), (1, 2), (2, 1)$ is not proved in Lean by this receipt. These three constant graphs do not require their existence decision for the ray theorem: any fixed graph has a constant decision. The saved finite mirror finds paths there, which is finite evidence outside Lean.

The specification and challenge were fixed before the proof by a separate statement writer. One independent AI statement reader returned "faithful with caveats", confirming the path domains and the carbon order $14t^2 + 16td + 2d^2$, while retaining a caveat about identification with the named catalogue family. Its 36 known pairs lie within the saved mirror's 80 pairs $1 \leq t \leq 8$,

$1 \leq d \leq 10$; they are not a second disjoint census or a universal catalogue-isomorphism theorem. The received mirror agrees with the verifier's construction on the recorded graphs and decisions. No new owner-confirmation field, human proof review or independent Lean derivation is added. The inert challenge has two intentional placeholders and is excluded from proof dependencies. Source and record hashes are supplied in the machine layer.

7.5. **Historical conditional kernels.** *Machine-readable claim:* `lem:historical-kernels`.

The earlier research layer supplies conditional pieces: `finite_literal_path_decision` assumes a finite type and two common neighbours; GC lattice transport excludes odd endpoints in a scaled lattice; the four cylinder arithmetic declarations prove the height-kernel displacement with stated divisibility hypotheses; `induction_uniform_depth` assumes a finitely branching child list and genuinely decreasing lexicographic rank. The concrete `ietComponentChildren` models only removal of equal last labels, not every subtraction, equal-length elimination or accelerated suffix. The geometric hypotheses, all-start coverage, full compressed action and a certified program are not supplied by those conditional types. There were historically 57 such research declarations; the former total 98 also counted 41 A3 arithmetic entries. Neither count is a formalization percentage or the current library size.

7.6. **Pins, receipt scope and proof dependencies.** The received toolchain is Lean 4.35.0 with the fixed mathlib manifest. The earlier arithmetic and K3 sufficiency reports record only `propext`, `Classical.choice`, `Quot.sound` as transitive axioms. Their receipt bytes and scopes are retained. For Theorem 1a, its separate source closure and the earlier research-team build/axiom output remain fixed. For the new K3 results, the coordinator records a successful build and standard-three-axiom queries for all three entry declarations. The research team receives and hash-checks these records without running Lean.

The saved six-kernel K3 Comparator receipt checks `ClaudeChallengeK3.main` in paranoid mode only; it is not a separate six-kernel check of `k3_statement` or `noThree`. It is also distinct from the earlier Theorem 1a and A3 receipts. All six kernels check the same proof, not six independent derivations. The K3 execution lacked OS process isolation on macOS. For C1, a separate paranoid Comparator receipt covers both `ClaudeChallengeC1.classification` and `ClaudeChallengeC1.main`: all six kernels accepted the same two supplied proofs. The coordinator also records a successful build and standard-three-axiom queries for both declarations. The worker report predates that Comparator run; its not-run note is retained as historical. C1 likewise lacked OS process isolation on macOS. A separate O2 paranoid Comparator receipt likewise covers both `ClaudeChallengeO2.classification` and `ClaudeChallengeO2.main`, with all six kernels accepting the same supplied proofs. The coordinator reports successful builds and standard-three-axiom queries for both. The worker report precedes the Comparator run and its not-run note is historical. O2 also lacked OS process isolation on macOS. No external kernel or Lean command is rerun for this edition. The claims' expository dependencies and Lean's imported proof-term dependencies are different. A prose statement can remain open for its coordinate or human-semantic bridge while a separate exact formal declaration is proved. Conversely the proved arithmetic predicates do not supply their graph iff interpretation. No human peer review of the proofs is recorded and novelty is not established.

8. An extension to fixed oblique directions (proof sketch)

Machine-readable claim: `claim:oblique`.

Status. The coordinator has reviewed the skeleton of AN Sections 2–6 as a plausible proof sketch. Explicit Section 5 transfer conditions and numerical validation remain incomplete; no independent second proof covers this wider domain. Four obligations below remain. This section is not part of the main theorem or the abstract's result.

The proposed extension replaces the six side directions by an arbitrary fixed finite set of primitive lattice directions, keeping eventual validity, ownership and fixed affine gluing. For a direction v, use transverse coordinate $h = \det(v, z)$. Across a seam the shift is affine. Let D be a common multiple of the nonzero $|\det(v, e)|$ and put $Q = 4D$. Add the transverse cell modulo Q and t modulo Q to the finite labels. Entrance and exit indices are $k_{\text{in}} = \lceil \xi_{\text{in}} \rceil$, $k_{\text{out}} = \lceil \xi_{\text{out}} \rceil$ with $k_{\text{in}} \le k < k_{\text{out}}$. Reordering cells $n = \sigma + Qq$ conjugates the discrete midpoint permutation; it does not claim an order-preserving conjugacy of the continuous intervals.

The AL/AM/AH proposal keeps a full partition into return towers, including saved periodic components. Equal final labels remove a periodic tower; equal lengths merge and remove a label without losing an orbit. Repeated winner cycles are compressed as powers. The claimed bound is on the number of Letter/Cat/Power nodes, not on expanded passage count. Base widths and exponents are affine on fixed residue branches; word lengths and displacement vectors are rational polynomials in t. A lexicographic measure uses leading coefficient sum in the original parameter, number of labels, and constant length; it is not silently reused after coefficient-amplifying reparameterization.

For local ownership, close the side directions under rotations. A unit triangle meets at most $\sum_e (\max_\Delta \det(e, z) - \min_\Delta \det(e, z) + 1)$ candidate macro-edge levels. Embedded noncrossing edges and finitely many corner displays bound the length of the local gluing word. A finite number of charts alone does not bound how often they can be revisited. Entrance-closed, exit-open ownership excludes repeated representations of one position from genuine quotient collisions. Keep the full corner class and the degree-six second-point test; non-IPR graphs can have fewer than 120 legal starts.

Tower return restores midpoint and direction; two returns restore left/right phase. Nonempty output, initial suffix and terminal prefix must be connected before using this as a finite first-repeat bound. Positions are leaf types and ancestor-power repetition indices. A free index j contributes j times a displacement depending only on t. Products jt or jt^2 are allowed coefficients in one-parameter Presburger arithmetic; products of two freely quantified repetition indices are not used. Fix one start address θ throughout:

$$\begin{aligned}\mathrm{Bad}(t, \theta, e) \iff \exists \xi, \eta\, [D_\theta(\xi) \wedge D_\theta(\eta) \wedge \xi \le e \wedge \eta \le e \\ \wedge \neg I_\theta(\xi, \eta) \wedge G_\theta(\xi, \eta)].\end{aligned}$$

Success would be $\exists \theta, e\, [\mathrm{Legal}(t, \theta, e) \wedge \neg \mathrm{Bad}(t, \theta, e)]$. Legal includes odd endpoint index, minimum length, degrees and the preceding output position. Allowing each collision a different start address would be incorrect.

The remaining obligations are: (i) write the first-crossed-edge, containment and order inequalities of the Section 5 transfer, with an oblique-family example; (ii) verify the AH/AM affine tower coefficients for the Q-split IET; (iii) keep effective threshold/period computation dependent on the effective elimination result [7], whose full proof was not reaudited; (iv) compare an oblique-family ray's predicted tail against full graph decisions. None is newly resolved here. BGW supplies eventual periodicity only after the geometric formula has been established.

8.1. **Prospective applications only.** C2 and K4 are specified leapfrogs of C1 and K3; the identified O1$(p, 0, s)$ subfamily is a specified leapfrog of O2. The local coordinate map $(x, y) \mapsto (x - y, x + 2y)$ has determinant 3. These are prospects through the extension, not families in Corollary 1b. Leapfrog is not assumed to preserve absence in general, and the full three-parameter O1 family is not classified. These prospects do not cover arbitrary signatures or every degenerate boundary. For the prospective signatures, twice area is $24a^2 + 48ab + 12b^2$ for C2 and $42r^2 + 36rs$ for K4. Their $a = 0$ and $r = 0$ cases collapse faces, while $b = 0$ and $s = 0$ give zero sides. The zero middle coordinate in the identified O1 subfamily denotes a separately identified leapfrog signature, not permission to discard a zero edge in the old signature.

9. Fixed-width structure, finite checks and novelty

9.1. **Belts and a fixed cylinder.** *Machine-readable claim:* `prop:cylinder`.

Developed triangle midpoints lie on a straight line; vertices alternate on its two neighbouring integer-height rows. Under $z' = Mz + q$, a height $h(z) = n \cdot z + e$ transforms as $n' = nM^{-1}$, $e' = e - n' \cdot q$. If every start is covered by a return preserving position, direction and primitive height, no cone lies inside a band, and all widths are at least two, the required pair of endpoint rows cannot both occur. Width one alone is not sufficient: simplicity and endpoint-neighbour degrees remain necessary. For integer $\mathrm{GC}(k, 0)$, $k \geq 2$, the odd-point determinant ± 1 contradicts membership in the k-multiple lattice; rotations about parent vertices preserve that scaled lattice.

For the fixed-cylinder argument AD let circumference vector $W = gu$, $g = \gcd(|W_1|, |W_2|)$, $h(z) = \det(u, z)$, and insertion vector D have $H = h(D) > 0$. Let L be the least common multiple of the nonzero $|h(v)|$ over the six unit directions. An entrance-state period is

$$Q = 2gL, \qquad \frac{Q}{h(v)}(Hv - h(v)D) \in \mathbb{Z}W. \tag{8}$$

Directions with $h(v) = 0$ loop in the flat core and repeat without reaching a cone. If the caps and collars have a fixed finite vertex set C, at most $|C| + 1$ new cap visits occur before stopping. Overlap of the same three-vertex display is not a new visit. On a residue branch the passages have form $a_i(n) + jv_i$, $0 \leq j \leq \ell_i(n)$, with affine a_i, ℓ_i and fixed v_i, W. All collisions are $a_i(n) + jv_i = a_k(n) + qv_k + zW$, with distinct-position and range conditions. Ordinary Presburger arithmetic then gives periodicity. Q is an entrance period, not necessarily the final or least period.

The uniform cap argument does not extend to growing width/caps. The old $\mathrm{D12}(3, 3t + 4)$ example has passage word $(BA)^{t+1}$ and first legal endpoint index $30t^2 + 88t + 59$. Its $2(t + 1)$ passages are unbounded. The former general bounded-passage forecast was withdrawn.

9.2. **Dated finite evidence.** *Machine-readable claim:* `sec:finite`.

The packet retains, rather than reruns, the relevant evidence. T0770 compares 1,809 arithmetic values on nine known A3 rays; T0771 adds fourteen graph decisions on an independent constructor. Those fourteen are not unseen tests and the 1,809 values are not counted twice. T0887 checks fixed tail coefficients beyond 64 on those rays. T0888 audits E10's fixed descriptor: eight charts, 36 inverse gluings, 108 intervals, 156 connections, 120 descriptors and twelve labels. T0889 audits 71 charts, 342 edges and 21 single-zero boundaries. These are finite checks of supplied data, not the general ownership theorem. The recorded AM audit has 6,564 inputs and 67,312 states; the AN phase audit has 48 directions, 14,904 entrances, 4,005 intervals (1,013 empty), 5,496 lifts and 40,380 conjugacy cells.

R004's report-14 finite arithmetic check has 378 coefficient representations and 13,430 pairs, maximum recorded period 44 and threshold 69. The report-15 checks include 7,056 strips, 2,255 nonempty row outputs, 16,212 subintervals, 99,225 ordered row pairs and 509,160 folded-point comparisons. Their hand-found row correction and earlier type/key failures are kept with their chronology. Numerical agreement does not replace either general proof or repair its hypotheses. No graph construction, R004 checker, or historical scientific audit was rerun for this translation.

The K3 worker's earlier finite mirror agreed with its specification and the verifier's construction on the stated known samples. The two constructions and generated Lean text are separate from the kernel proof. Generated degree formulas were fitted from sample data, but the received Lean theorem checks their universal statements; samples alone would not justify sufficiency. The worker's mirror is not a new independent reviewer of its own specification.

9.3. **Novelty and remaining obligations.** *Machine-readable claim:* `sec:novelty`.

Immediate known consequences. Once a fixed sentence ψ_η describes success on a^n, $t = 2n + \eta$, MSO/regular-language equivalence gives a finite automaton; its unary state orbit is eventually a cycle [5, Section 3.1]. The HTML numbering is Theorem 3.2; the same v1 PDF calls it Theorem

14. Finite fixed-modulus residue branches preserve eventual periodicity. Likewise, a fixed-size parametric Presburger description gives the conclusion directly by [6, Theorem 1.15]. Neither theorem builds the geometric description: an unbounded list of path vertices is not a fixed list of quantified variables, and variable-variable products are not permitted by that Presburger theorem.

For a fixed-width construction $G_n = C[F^n(B)]$ with fixed finite boundary, literal-path existence has a fixed CMSO_2 description. Quantify a vertex set X, an edge set E' and distinct endpoints a, b: (X, E') is connected, has degree one at a, b and degree two elsewhere; $|X| \geq 4$ is even; a, b have ambient degree five and their path neighbours ambient degree six; every two-edge path in E' has adjacent outer vertices in the ambient graph. This permits interior cones and extra chords. The selected subgraph is precisely a literal path, in both directions. Finite-state recognizability for a fixed HR signature [9, Chapter 1, Theorems 1.12 and 1.21(2)] then makes iteration of F a finite-state orbit. Thus this fixed-width existence statement is a short known consequence, not a new principle. Mere bounded treewidth of an arbitrary indexed sequence is insufficient: the uniform repeated construction is essential. The explicit Q, cap-visit bound and actual period-calculation procedure require separate evaluation.

Nearby results and the missing translations. Behmaram–Boutillier's fixed-circumference matching transfer matrix has 2^m states [10, Section 3, Eq. (3.1)]; increasing width does not leave the same matrix, and its observable is matchings, not literal paths. Fischer–Makowsky prove periodicity or semilinearity of model-size spectra for CMSOL classes with bounded width [11, Theorems 1.4, 1.7 and 1.10]. Such a spectrum asserts existence of some model of a size, rather than deciding a specified G_t. Kotek–Makowsky's labelled (bi-)iterative graph families and MSO polynomials admit finite matrix transfers and C^2-finite recurrences [12, Definitions 17–20, Lemma 28, Theorems 30 and 32]. Their bounded-clique-width framework does not automatically include growing lattice patches; a recurrence for values is not itself Boolean periodicity. In each case the geometric family and its parameter require a uniform interpretation satisfying the cited hypotheses.

The syntax $\pm x \pm y \leq c$ is already the octagon domain [13, p. 1]. Bozga–Iosif–Konečný's difference, octagonal and finite-monoid affine relations have acceleration and Presburger transitive-closure results [14, Definition 1 and Sections 4.2.1–4.2.2]. These principally iterate a fixed relation and allow linear drift; here the relation itself varies with t, and all previously output surface vertices must remain distinct. A fixed-dimensional translation preserving that condition has not been supplied. The term "zigzag automata" there does not identify the subject with fullerene zigzags.

Jonoska–Krajcevski–McColm study counter languages of walks returning to a unit cell in periodic digraphs [15]; Wanke studies path/cycle algorithms on finite periodic graphs [16]. For these two sources (S2191, S2192), only the official abstracts and bibliographic records were obtained. We do not infer their full hypotheses or claim an application from an unread theorem. Walks or decidability alone do not preserve literal simplicity or prove periodicity of the parameter set.

Unary length is essential. The standard WS1S encoding of Presburger integers uses their binary representations [17, printed pp. 11–12]. Binary regularity does not imply eventual periodicity of the represented integers: the regular language 10^* represents powers of two. Our reduction uses n as unary word length, not the binary digits of t. A separate formula for each t would also be insufficient.

Boundary of the contribution. The received R005 comparison read the theorem statements and specified sections listed in the packet's source records, not all proofs. Within that reading, no prior result giving the full claim from the same geometric hypotheses was located. Novelty is not established. The remaining candidate contribution is the uniform, two-way translation of geometric passages, seam ownership, legal truncation and all surface-vertex collisions into one fixed MSO sentence, even with growing widths and passage counts. Fixed-width theory and the known logical conclusions do not supply that translation by themselves; we do not prove that a shorter existing reduction is impossible. The general convex-face proof is received research work, not an already published theorem cited as external authority. Human proof peer review and full comparison with

catalogue/zigzag theory remain necessary. The extension's unfinished geometric details do not disappear on receipt of the convex-face proof.

The two-interval arithmetic kernel is covered by Li–Chen's uniform bound for continued fractions of ratios of parameter polynomials [8, Theorem 27]. For several intervals, finite residue branches and a uniform compressed node count, the stored survey did not locate an identical general theorem; this is not a novelty proof.

K3's variable-modulus behaviour obstructs a single ordinary multivariable Presburger description in fixed common coordinates. Moving to canonical representatives or allowing variable divisibility does not follow from raywise periodicity. The stronger M4/M4-prime questions remain separate. General E10 $r = 1$ necessity, untranscribed families, all degenerations, human proof peer review, and a formal geometric-to-program pipeline are not completed here. A3 minimum, density and the IPR census belong to the first article. No new D12 classification is asserted by this paper.

Appendix A. Local polygons and zero boundaries

Machine-readable claim: `app:domains`.

Twice total polygon area is the carbon order only when the quotient is valid. The stored local coefficient audit gives

Signature	Side factors	Twice total area
E10	$p, r, r+s$	$12r^2 + 2s^2 + 12rs + 24pr + 12ps$
D12	$s, r+s$	$12r^2 + 48rs + 48s^2$
D14	r, s	$8r^2 + 24rs + 12s^2$
D1	p, r	$36p^2 + 48pr + 12r^2$
K3	r, s	$14r^2 + 12rs$
B2	p, r	$12p^2 + 24pr + 4r^2$
C1	a, b	$8a^2 + 16ab + 4b^2$
O2	t, d	$14t^2 + 16td + 2d^2$

The 71-chart/342-edge audit establishes local convexity, positive area and side lengths in its stated positive domain, not global quotient simplicity. With other parameters positive, E10 $s = 0$ and D12 $r = 0$ alone remain locally nondegenerate. E10 $p = 0$ or $r = 0$, K3 $r = 0$, B2 $p = 0$ or $r = 0$, C1 $a = 0$, and O2 $t = 0$ collapse some faces. D12 $s = 0$, either D14/D1 zero, K3 $s = 0$, C1 $b = 0$ and O2 $d = 0$ produce zero sides. Removing an IET zero-length label is not the same as redefining a graph after cone mergers. The omitted leapfrog-domain data remain in the extension's prospective records, not the main corollary. A3's $l = 0$ and $k = l$ are the separate A1 and A2 boundaries.

Appendix B. Representative formal statements and source correspondence

The accompanying `v3-correspondence.md` maps every numbered v3 section to this reorganization, including the moved oblique-direction sketch. It distinguishes translation, condensation and formal-status updates; no prose-equivalence theorem is claimed.

The primary received declarations now include `ClaudeChallenge2.main` and `ClaudeChallenge2.State.main_statement` and `ClaudeChallenge2.main_residue`, followed by `Claude2.k3_ray_all`, `Claude2.e10_ray_all`, `Claude2.k3_example`, `Claude2.e10_example`, and `Claude2K3.k3_sufficient`. The new K3 entries are `ClaudeChallengeK3.main`, `Claude4K3.k3_statement` and `Claude4K3.noThree`. The new C1 entries are `ClaudeChallengeC1.classification` and `ClaudeChallengeC1.main`. The O2 entries are `ClaudeChallengeO2.classification` and `ClaudeChallengeO2.main`. The machine layer supplies their exact types, source commit and the kind of proof receipt. The historical representative kernel types remain:

> `finite_literal_path_decision`: finite type, two-common-neighbour hypothesis, cone predicate and fixed oriented triangle imply a first repeat at most the number of vertices, and path existence iff a legal endpoint occurs before it.

`cylinder_portal_shift`: a Bézout witness for primitive u and $g \mid Q/\mathrm{cylHeight}(u, v)$ imply that the stated shifted displacement is a multiple of gu. The integer-division and divisibility hypotheses are retained.

`induction_uniform_depth`: a finite-list child relation with a strictly decreasing lexicographic natural rank has a finite uniform depth from each starting state.

`iet_component_uniform_depth`: the concrete equal-final-label removal component has such a depth; the omitted IET branches are not thereby formalized.

The supplied exact sources, not these paraphrases, determine the Lean types. Dependency revisions are pinned in `lake-manifest.json` and the toolchain in `lean-toolchain`. Caches, compiled binaries and raw graph streams are intentionally omitted; no separate artifact collection is supplied. For optional local reproduction, obtain the pinned dependencies through Lake and use `lake exe cache get` for mathlib build artifacts. Contact is through GitHub Issues at the repository specified in `REPOSITORY-URL.txt`.

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